PRD — Dental Clinic Management Platform

1. Overview

Product name: DentaFlow
Version: 1.0 (MVP)
Owner: Clinic Product Owner
Status: Draft

DentaFlow is a cloud-based clinic management platform for dental practices. It helps front-desk staff, dentists, and hygienists manage patients, appointments, treatment records, billing, and reminders in one place.

2. Problem statement

Dental clinics often manage appointments, patient histories, consent forms, treatment plans, invoices, and follow-ups across disconnected tools such as paper files, spreadsheets, phone calls, and generic calendar apps.

This creates:

- Missed or double-booked appointments
- Incomplete medical and dental histories
- Slow insurance and payment workflows
- Poor patient follow-up and recall
- Difficulty tracking chair utilization and revenue

3. Goals and objectives

Goal	Objective
Reduce no-shows	Send automated appointment confirmations and reminders
Improve care quality	Maintain complete, searchable patient and treatment records
Increase operational efficiency	Digitize scheduling, check-in, billing, and follow-up
Improve revenue visibility	Track treatments, payments, outstanding balances, and chair utilization
Support compliance	Keep clinical documentation, consent, and audit history organized

4. Target users

User	Primary needs
Patient	Book, reschedule, or cancel appointments; receive reminders; view treatment and payment information
Front-desk staff	Manage schedule, check-ins, patient intake, payments, and communications
Dentist	View medical/dental history, chart treatments, create treatment plans, prescribe, and document care
Dental hygienist	Record cleanings, assessments, notes, and follow-up recommendations
Clinic manager/owner	Monitor KPIs, staff schedules, revenue, inventory, and patient growth
Billing/admin staff	Manage invoices, insurance claims, payment status, and refunds

5. Scope

In scope for MVP

- Patient registration and profile management
- Appointment scheduling and calendar
- Automated SMS/email appointment reminders
- Patient intake forms: medical history, dental history, allergies, medications, consent
- Clinical notes and treatment records
- Treatment plan creation and procedure catalog
- Invoicing, payments, and outstanding-balance tracking
- Basic reporting dashboard
- Role-based access control
- Audit trail for record changes

Out of scope for MVP

- Full insurance claim adjudication
- AI-based diagnosis
- Tele-dentistry/video consultations
- Multi-clinic franchise management
- Native mobile apps
- Advanced inventory forecasting

6. Key features and requirements

6.1 Patient management

User story: As a front-desk staff member, I want to create and update patient profiles so that the clinic has accurate contact, medical, and dental information.

Requirements:

- Create patient records with name, date of birth, gender, contact details, address, emergency contact, and insurance details
- Capture medical history, dental history, allergies, medications, and consent status
- Support document uploads: X-rays, referral letters, ID, signed forms
- Search patients by name, phone, email, or patient ID
- Maintain a chronological timeline of visits, treatments, notes, and communications

6.2 Appointment scheduling

User story: As a patient, I want to book an appointment online so that I can choose a convenient time without calling the clinic.

Requirements:

- Calendar view by day, week, and dentist/chair
- Book, reschedule, cancel, and waitlist appointments
- Appointment types: new patient, cleaning, filling, root canal, extraction, orthodontics, emergency
- Assign dentist, hygienist, chair/room, and duration
- Prevent double-booking
- Support recurring appointments for multi-visit treatments
- Online booking portal with available slots
- Automated confirmation, reminder, and cancellation messages

6.3 Clinical workflow

User story: As a dentist, I want to review a patient's history and document treatment so that care is consistent and auditable.

Requirements:

- Pre-visit summary: allergies, medications, prior treatments, notes, alerts
- Dental chart with tooth-level status and procedure mapping
- Clinical notes with templates for exams, cleanings, procedures, and follow-ups
- Treatment plan with proposed procedures, costs, alternatives, and patient approval

- Attach images, X-rays, and lab documents
- Record diagnosis, procedures performed, materials used, and next-step recommendations
- Prescription recording with dosage, frequency, and duration

6.4 Billing and payments

User story: As a billing staff member, I want to generate invoices and track payments so that the clinic can reduce outstanding balances.

Requirements:

- Generate estimates and invoices from treatment plans
- Apply discounts, taxes, and payment plans
- Accept cash, card, UPI/bank transfer, and insurance reference
- Track paid, partial, pending, refunded, and written-off payments
- Insurance claim status tracking
- Printable and emailed receipts
- Daily cash-up and payment reconciliation

6.5 Patient communication

Requirements:

- Appointment reminders via SMS/email
- Recall reminders for cleanings and follow-ups
- Post-treatment instructions
- Feedback/review request messages
- Communication log visible on the patient profile

6.6 Reporting and analytics

User story: As a clinic owner, I want to see operational and financial KPIs so that I can improve performance.

MVP metrics:

- Appointments booked, completed, cancelled, and no-show rate
- New vs. returning patients
- Revenue by service, dentist, and period
- Outstanding patient balances
- Chair utilization
- Treatment-plan conversion rate
- Recall compliance rate

7. Non-functional requirements

Area	Requirement
Availability	99.5% uptime target during clinic hours
Performance	Patient search results within 2 seconds for typical clinic data volume
Security	Encryption in transit and at rest; role-based access; session timeout
Privacy	Patient data access limited to authorized roles; full audit trail
Auditability	Log create, update, delete, view-sensitive-record, and billing actions
Backup	Daily automated backups with restore capability
Accessibility	WCAG 2.1 AA target for patient-facing screens
Localization	Support clinic currency, date format, language, and timezone
Integrations	SMS/email provider; payment gateway; calendar export; optional accounting integration

8. User flows

New patient booking

1. Patient opens online booking page.
2. Patient selects service, dentist, date, and time.
3. Patient enters contact and basic intake details.
4. System checks availability and creates appointment.
5. Patient receives confirmation and intake-form link.
6. Front desk verifies details on arrival and checks the patient in.

Treatment visit

1. Dentist opens patient chart and reviews alerts/history.
2. Dentist performs examination and records findings.
3. Dentist creates or updates a treatment plan.
4. Patient approves the plan and cost estimate.
5. Dentist documents completed procedures.
6. Billing generates invoice and records payment.
7. System schedules follow-up or recall.

9. Success metrics

Metric	Target
No-show rate	Reduce by 25% within 3 months
Online booking share	30% of appointments within 6 months
Average appointment scheduling time	Under 60 seconds
Treatment-plan conversion	Increase by 15%
Outstanding balance aging	Reduce balances older than 60 days by 20%
Patient record completeness	95% of active patients have updated medical/dental history
Staff adoption	90% of daily appointments managed in the platform

10. Assumptions

- The clinic has stable internet connectivity.
- Staff will be trained before go-live.
- Existing patient records can be imported through CSV or manual entry.
- Payment gateway and SMS/email providers will be selected during implementation.
- Clinical records remain under dentist review and responsibility.

11. Risks and mitigations

Risk	Mitigation
Staff resistance to new workflow	Role-based training, simple UI, phased rollout
Data migration errors	Validation reports, sample migration, backup before cutover
Missed reminders due to wrong contact data	Require contact verification at check-in
Privacy/compliance exposure	Access controls, audit logs, data-retention policy
Low patient portal adoption	Simple booking flow, QR code at reception, reminder links

12. Release plan

Phase	Duration	Deliverables
Phase 1	Weeks 1–4	Patient profiles, scheduling, reminders
Phase 2	Weeks 5–8	Intake forms, clinical notes, dental chart
Phase 3	Weeks 9–12	Treatment plans, billing, payments
Phase 4	Weeks 13–16	Reports, audit trail, integrations, go-live

13. Open questions

- Which countries and insurance providers must be supported first?
- Should patients be able to message dentists directly?
- What is the preferred payment gateway?
- Should X-ray imaging integrate with an existing PACS/dental imaging system?
- What data-retention period is required by local regulations?